

# Adaptive State-of-Health Forecasting for Second-Life LFP Cells Using Physics-Informed Synthetic Data and Hybrid Model Selection

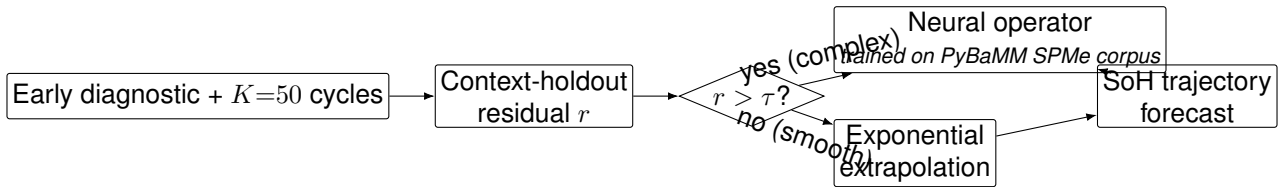
Krishna Chaitanya Vaddepally

Turno (Blubble Pvt. Ltd.), Bengaluru, India • krishna.vaddepally@turno.club

Theme: Data-driven Methods • Presentation: Oral (poster acceptable)

Second-life reuse of retired lithium iron phosphate (LFP) cells requires fast prediction of future degradation for each incoming cell. Instead of assuming a single degradation model is optimal for all cells, we develop an adaptive forecasting framework that selects between simple empirical extrapolation and a PyBaMM-trained neural operator based on early-cycle degradation complexity.

**Approach (Figure 1).** For each target cell the pipeline uses: (1) a short diagnostic characterisation together with the first  $K=50$  operating cycles as early-cycle context; (2) a PyBaMM [1] SPMe corpus on Prada2013 LFP chemistry [2] with perturbed degradation trajectories around seven anchor cells from three suppliers (CALB, EVE, REPT), providing training trajectories that cover the second-life SoH range; (3) a  $\theta$ -conditioned Neural ODE [3] operator that forecasts the residual SoH trajectory from the diagnostic features, the  $K=50$  context window and degradation-model-derived features; (4) an exponential  $SoH(n)=a+be^{-cn}$  baseline fitted on the same context window; and (5) a residual-based selector that predicts the last 20% of the context from a fit on the first 80% and picks the neural operator whenever the context-holdout residual exceeds a predeclared threshold  $\tau$ , chosen on a synthetic-only selector pool disjoint from the external cells.



**Figure 1:** Adaptive selector between empirical exponential and physics-informed neural forecast. A context-holdout residual measures early-cycle degradation complexity; the predeclared threshold  $\tau$  is chosen on synthetic trajectories, not on the external cells.

**Results.** Across three external LFP cells excluded from operator training (CALB\_0029, EVE\_0003, REPT\_0028), exponential forecasting achieved 0.70 pp mean RMSE, while the physics-informed operator achieved 0.80 pp. A context-residual selector combined the complementary models and reduced mean RMSE to 0.47 pp, matching the oracle model selection on the evaluated cells. The residual-based selector reproduced the oracle model choice on the three evaluated external cells: exponential for the two smoother trajectories (CALB\_0029, REPT\_0028) and the neural operator for the noisier EVE\_0003. The exponential baseline and the operator show complementary strengths – exponential wins on smooth fade while the operator wins on the more nonlinear trajectory – motivating selection rather than committing to either model alone.

**Table 1:** External held-out SoH RMSE (percentage points) on three cells excluded from operator training. The hybrid row uses the full- $\theta$  operator when the residual-based selector chooses it.

| Method                     | CALB_0029   | EVE_0003    | REPT_0028   | Mean        |
|----------------------------|-------------|-------------|-------------|-------------|
| Exponential                | 0.44        | 1.54        | 0.13        | 0.70        |
| Full- $\theta$ operator    | 1.06        | 0.84        | 0.51        | 0.80        |
| No- $\theta$ operator      | 1.08        | 0.89        | 0.49        | 0.82        |
| Reduced- $\theta$ operator | 1.18        | 0.97        | 0.55        | 0.90        |
| Hybrid ( $\tau=0.54$ pp)   | <b>0.44</b> | <b>0.84</b> | <b>0.13</b> | <b>0.47</b> |

Ablations at matched split and seed indicate that PyBaMM-generated trajectories improve training diversity, while inference performance is primarily driven by early-cycle degradation behaviour rather than explicit degradation-parameter conditioning.

The current evaluation is limited to three external cells, and longer-horizon projections require additional experimental validation.

[1] V. Sulzer et al., *J. Open Res. Software* **9** (2021), 14. [2] M. Prada et al., *J. Power Sources* **219** (2012), 353–362.

[3] R. T. Q. Chen et al., *NeurIPS* (2018).